

E07 Adiabatic Batch Reactor Example

The irreversible, elementary liquid phase reaction $A + B \rightarrow C$ is carried out adiabatically in a 10-L batch reactor. Species A and B are present initially in equimolar amounts. Determine the time needed to achieve 90% conversion and the resulting temperature. $T_0 = 300\text{ K}$.

$$C_{A0} = 0.1\text{ mol/L}$$

$$C_{PA} = C_{PB} = 0.015\text{ kcal/mol/K}$$

$$C_{PC} = 0.030\text{ kcal/mol/K}$$

$$\Delta H_{f,A}^0 = -20\text{ kcal/mol}$$

$$\Delta H_{f,B}^0 = -15\text{ kcal/mol}$$

$$\Delta H_{f,C}^0 = -41\text{ kcal/mol}$$

$$E_{act} = 10\text{ kcal/mol}$$

$$k = 0.01\text{ L/mol/s at } 325\text{ K}$$

mole balance $\circ$ $N_{A0} \frac{dX}{dt} = -r_A V$

liquid phase so V is constant, so

can simplify to $C_{A0} \frac{dX}{dt} = -r_A$, $C_{A0} = 0.1 \frac{\text{mol}}{\text{L}}$

rate law $\circ$ $-r_A = k C_A C_B$

concentration eqs $\circ$

$$C_A = C_B = C_{A0} (1-X)$$

$$\circ \circ \quad -r_A = k C_{A0}^2 (1-X)^2$$

Arrhenius eq $\circ$

$$k = \left(0.01 \frac{\text{L}}{\text{mol} \cdot \text{s}} \right) \exp \left[\left(\frac{10 \frac{\text{kJ}}{\text{mol}}}{0.001986 \frac{\text{kJ}}{\text{mol} \cdot \text{K}}} \right) \left(\frac{1}{325 \text{ K}} - \frac{1}{T} \right) \right]$$

Energy balance:

2 ways, either is fine:

$$\textcircled{1} \quad T = \frac{T_0 \sum \Theta_i C_{p,i} + X(-\Delta H_{rxn} + \Delta C_p \cdot 298.15K)}{\sum \Theta_i C_{p,i} + X \Delta C_p}$$

terms:

$$\rightarrow \Delta H_{rxn} = \Delta H_{f,c}^\circ - \Delta H_{f,A}^\circ - \Delta H_{f,B}^\circ = -6 \frac{\text{kcal}}{\text{mol}}$$

$$\rightarrow \sum \Theta_i C_{p,i} = \Theta_A C_{p,A} + \Theta_B C_{p,B} + \Theta_C C_{p,B}$$

$$= (1) \left(0.015 \frac{\text{kcal}}{\text{mol} \cdot K} \right) + (1) \left(0.015 \frac{\text{kcal}}{\text{mol} \cdot K} \right) + (0) \left(0.03 \frac{\text{kcal}}{\text{mol} \cdot K} \right)$$

$$= 0.030 \frac{\text{kcal}}{\text{mol} \cdot K}$$

$$\rightarrow \Delta C_p = \sum_{\text{products}} \left| \frac{\nu_i}{\nu_A} \right| C_{p,i} - \sum_{\text{reactants}} \left| \frac{\nu_i}{\nu_A} \right| C_{p,i}$$

$$= C_{pC} - C_{pA} - C_{pB} =$$

$$= 0.030 \frac{\text{kcal}}{\text{mol} \cdot \text{K}} - 0.015 \frac{\text{kcal}}{\text{mol} \cdot \text{K}} - 0.015 \frac{\text{kcal}}{\text{mol} \cdot \text{K}}$$

$$\Delta C_p = 0$$

$$(300\text{K})(0.030 \frac{\text{kcal}}{\text{mol} \cdot \text{K}}) + X (6 \frac{\text{kcal}}{\text{mol}})$$

$$\therefore T = \frac{0.030 \frac{\text{kcal}}{\text{mol} \cdot \text{K}}}{0.030 \frac{\text{kcal}}{\text{mol} \cdot \text{K}}}$$

- solve simultaneously

way ②

use the general ER:

$$\frac{dT}{dt} = \frac{(\Delta H_{rxn})(r_A V)}{N_{A0} (\sum \theta_i C_{pi})} = \frac{(-6 \frac{\text{kcal}}{\text{mol}})(r_A)}{C_{A0} (0.030 \frac{\text{kcal}}{\text{mol} \cdot \text{K}})}$$

- where $C_{A0} = 0.1 \frac{\text{mol}}{\text{L}}$ and

$$r_A = -k C_{A0}^2 (1-X)^2 \quad \text{and}$$

k is the function of T from above

couple the EB w/ the MB

& solve simultaneously

NOTE: Since X & T both vary with

time (t), you must solve simultaneously.

PFR & batch reactor require

simultaneous solution even for an

adiabatic reactor, CSTR if you

have an adiabatic reactor, you

can solve sequentially (e.g. given X ,

solve for T , use T to get k (and K_c , C_i ,

etc as needed), use those to solve
your mole balance.)

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